Projeto de Filtro seletivo em Frequência: Parte 1

Lucila Gabriela,
Hanna Vitória,
Maria Luiza Melo.

Parâmetros 4: $f_1 = 5,2\,KHz$, $f_2 = 5,5\,KHz$,
$\delta_2 = 0,001$, $\Delta\omega = 0,04\,\pi rad$

· $A_r = 20 \log \delta'$

$A_r = 20 \log (0,001)$

$A_r = -60\,dB$ ⇝ Janela de Blackman, pois o dB dela é -74
e o de Hamming é -53.

· $f_a \geq 2B$

$f_a = 12\,KHz$

(*) Blackman:

$\Delta f = \dfrac{5,5}{M}$

$M = \dfrac{5,5}{\Delta f} = \dfrac{5,5}{0,02}$

$M = 275$

· $\Delta\omega = 2\pi \Delta f_d$

$0,04\,\pi rad = 2\pi \Delta f_d$

$\Delta f_d = \dfrac{0,04\pi}{2\pi} = 0,02$

· $\Delta f_d = \Delta f_{Hz} \cdot T$

$\Delta f_{Hz} = \Delta f_d \cdot f_a$

$\Delta f_{Hz} = 0,02 \cdot 12 \times 10^3$

$\Delta f_{Hz} = 240\,Hz$

$f_s = 5.000\,Hz$

· $\omega_s = 2\pi \dfrac{f_s}{f_a}$ ∴ $\omega_s = 2\pi \cdot \dfrac{5KHz}{12KHz} = 0,833\,\pi rad$

$\Delta\omega = \omega_s - \omega_p \Rightarrow \omega_p = \omega_s - \Delta\omega$

$\omega_p = 0,833\pi - 0,04\pi$

$\omega_p = 0,793\,\pi rad$

· $\omega_c = \dfrac{\omega_s + \omega_p}{2} = \dfrac{0,833\pi + 0,793\pi}{2} = 0,813\,\pi rad$

$$h[n] = \frac{\text{sen}(0{,}813\pi n)}{\pi n} \cdot \left(0{,}42 - 0{,}5\cos\left(\frac{2\pi}{275}\right)\right), \quad 0 \leq n \leq 275$$

$$w[n] = \begin{cases} 0{,}42 - 0{,}5\cos\left(\frac{2\pi}{275}\right) + 0{,}08 \cdot \cos\left(\frac{4\pi n}{275}\right), & 0 \leq n \leq 275 \\ 0, & \text{caso contrário} \end{cases}$$